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United States Patent	12409395
Kind Code	B2
Date of Patent	September 09, 2025
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Construction kit and components thereof

Abstract

A construction kit enables building of complex structures. The kit can include pole connectors, along with a series of hubs. For instance, two-point hubs, four-point hubs, eight-point hubs, five-point hubs, six-point hubs, and a clip can be the constituent parts of the kit. Some of the hubs can be flexible to provide further utility and additional angles for building. Complex structures, such as a ball-shaped connector can also be achieved from the base components.

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Appl. No.: 18/045808

Filed: October 11, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230114267 A1	Apr. 13, 2023

Related U.S. Application Data

us-provisional-application US 63254524 20211011

Publication Classification

Int. Cl.: A63H33/08 (20060101); A63H33/04 (20060101); A63H33/06 (20060101); A63H33/10 (20060101)

U.S. Cl.:

CPC **A63H33/044** (20130101); **A63H33/067** (20130101); **A63H33/088** (20130101);
A63H33/10 (20130101); **A63H33/106** (20130101); **A63H33/108** (20130101);

Field of Classification Search

CPC: A63H (33/067); A63H (33/088); A63H (33/106); A63H (33/108)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/254,524 filed Oct. 11, 2021, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to construction kits and building toys, and, specifically, to connectors useful in fort-building kits.

BACKGROUND OF THE INVENTION

(2) Toy construction kits are extremely popular and cover a wide gamut of different products. Children and/or hobbyists can construct a large variety of structures and models. Building bricks, notched sticks, magnetic connectors as well as more permanent arrangements involving adhesives are known in the art. However, these examples all face limitations based on their method of assembly. Frequently, the linkage of individual parts is limited to one or two configurations. As a result, the structures that can be built with such toys are often limited to arranging parts in parallel (i.e., linearly), and frequently the angles formed between individual parts are limited, sometimes only allowing for right angles.

SUMMARY OF THE INVENTION

(3) The present invention involves a construction kit for building forts. In an embodiment, the basic components of kits made in accordance with an embodiment of the present invention include connector hub pieces (“hubs”) and connecting poles (“poles”). The kit includes unique and versatile components enabling the assembly of a greater variety of constructed structures by giving the builder a large choice of options in terms of orientation of individual poles. Component parts with a large number of possible receiving ports offer additional flexibility both in terms of angles and the number of connectors that can be received at a single hub, leading to more complex building possibilities.

(4) In a further embodiment, the kit features hubs that can connect with each other, essentially turning certain hubs into adapters for others. When operating as adapters, the combined hubs can interface with other kit components in modified configurations. These adapters can enable the receipt of further poles, allow for more choices of angles between poles, or expand a constructed structure.

(5) A first component part of the present invention is an eight-sided hub. This piece includes a series of eight ports forming a ring around and circumscribing a pair of ports normal to the ring. In some embodiments the pair of ports have a dead-end construction, wherein each slot can individually receive a connector. In alternate embodiments, the two normal ports are interconnected, whereby they form a channel, enabling connector hubs or other pieces to pass through the channel, entering through one normal port and exiting out the other normal port. In such configurations, the eight-sided junction piece can receive connector hubs at eight different angles, or ten, if the normal ports are adapted to receive poles.

(6) Connector poles serve to connect joint pieces (i.e., hubs) to one another. The poles can be substantially or completely linear, operating to visually define a constructed structure. In an embodiment, the connector poles can have one or more closed ends. Such closed ends would be compatible with the ports of the hubs and sized and shaped appropriately to fit loosely or securely in the ports. In alternate embodiments, the connector hubs have open ends. This could be an innocuous design choice, or, alternatively, the open ends can enable the poles to interface with an adapter, a retention device, etc.

(7) Such adapters can fit loosely or securely in the connector hubs, using ridges, male-female connections, screw arrangements, etc. such that connections between the connector hubs and the joint pieces are made possible, or simply more secure. In other embodiments, two connector hubs can be connected via an adapter.

(8) In additional embodiments a hub in the shape of a ring is provided. Eight ports are placed along the circumference of the ring. In the center of the ring, there is provided an opening through which a connector hub piece or other component can pass through. In an embodiment, a pole and/or hub can be threaded through the opening in the ring. In certain embodiments, there is a mechanism for securing an adapter in place. A snap-fit, magnetic connection or other reversible connection can be utilized. In an exemplary embodiment, the adapter can be rotated clockwise or counterclockwise when centered in the opening of the ring to lock in the adapter. The adapter and ring can be securely coupled to one another through slots and/or interfitting ridges, pieces etc. which can be aligned through rotation of the adapter member.

- (9) In an embodiment, one or more hemispherical hubs can be provided that function as adapters. For instance, a hemispherical adapter can have a central hemispherical port configured to receive one end of a hub. In one embodiment, the hemispherical adapter has additional ports, which can be used to receive poles at additional angles. In certain embodiments, with hemispherical adapters in place on both ends of the adapter, a compound ball structure can be formed. In an embodiment, a total of eighteen different angles can be achieved for interfacing connector poles with the ball structure. The ball structure itself can serve as a compound joint piece or hub.
- (10) An advantage of kits made in accordance with embodiments of the present invention is the ability to assemble compound hub structures based on the needs of the builder. For instance, the ball structure could be partially assembled and still function.
- (11) In an exemplary embodiment, a two-point hub can serve as the base of a compound hub. The two-point hub, in its simplest form, can receive two poles and operate as purely a joint between two collinear poles. If additional ports are required, an eight-point hub can be coupled to the two-point hub and locked in place as described hereinabove. This would allow for placement of poles eight additional directions perpendicular to the ports of the two-point hub. From there, if the builder wants to place additional poles, or orient the poles to extend at additional angles from the combination of the two and eight-point hubs, one or two four-point hubs can be attached to provide four and eight additional angles to poles, respectively, and/or form the compound ball structure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present invention, reference is made to the following detailed description of various exemplary embodiments considered in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a schematic diagram showing a fort built using the building components of the present invention;
- (3) FIG. 2 is a schematic diagram showing an alternate view of the fort of FIG. 1;
- (4) FIG. 3 is a schematic diagram illustrating various components of the present invention in operation;
- (5) FIG. 4 is a perspective view of a pole connector in accordance with an embodiment of the present invention;
- (6) FIG. 5 is a front elevational view of the pole connector of FIG. 4;
- (7) FIG. 6 is a side elevational view of the pole connector of FIGS. 5 and 6;
- (8) FIG. 7 is a perspective view of a pole connector in accordance with an embodiment of the present invention;
- (9) FIG. 8 is a front elevational view of the pole connector of FIG. 7;
- (10) FIG. 9 is a side elevational view of the pole connector of FIGS. 7 and 8;
- (11) FIG. 10 is a perspective view of a five-point hub piece in accordance with an embodiment of the present invention;
- (12) FIG. 11 is a perspective view of a five-point hub piece in a flexed position, in accordance with an embodiment of the present invention;
- (13) FIG. 12 is a top view of a five point-hub piece in accordance with an embodiment of the present invention;
- (14) FIG. 13 is a top view of a five point-hub piece in a flexed position in accordance with an embodiment of the present invention;
- (15) FIG. 14 is an elevational view of a five-point hub piece in accordance with an embodiment of the present invention;
- (16) FIG. 15 is an elevational view of a five point-hub piece in a flexed position in accordance with

an embodiment of the present invention;

(17) FIG. **16** is a perspective view of a six-point hub piece in accordance with an embodiment of the present invention;

(18) FIG. **17** is a top view of a six-point hub piece in accordance with an embodiment of the present invention;

(19) FIG. **18** is an elevational view of a six-point hub piece in accordance with an embodiment of the present invention;

(20) FIG. **19** is a perspective view of an eight-point hub piece, in accordance with an embodiment of the present invention;

(21) FIG. **20** is an elevational view of an eight-point hub piece, in accordance with an embodiment of the present invention;

(22) FIG. **21** is a bottom view of an eight-point hub piece, in accordance with an embodiment of the present invention;

(23) FIG. **22** is a top view of an eight-point hub piece, in accordance with an embodiment of the present invention;

(24) FIG. **23** is a top perspective view of a two-point hub piece in accordance with an embodiment of the present invention;

(25) FIG. **24** is a front elevational view of a two-point hub piece in accordance with an embodiment of the present invention;

(26) FIG. **25** is a side elevational view of a two-point hub piece in accordance with an embodiment of the present invention;

(27) FIG. **26** is a top view of a four-point hemispherical hub piece, in accordance with an embodiment of the present invention;

(28) FIG. **27** is a side elevational view of a four-point hemispherical hub piece, in accordance with an embodiment of the present invention;

(29) FIG. **28** is a top view of a four-point hemispherical hub piece, in accordance with an embodiment of the present invention;

(30) FIG. **29** is a bottom view of a four-point hemispherical hub piece, in accordance with an embodiment of the present invention;

(31) FIG. **30** is a top perspective view of a composite structure in accordance with an embodiment of the present invention;

(32) FIG. **31** is a front elevational view of a different composite structure, in accordance with an embodiment of the present invention;

(33) FIG. **32** is a top perspective view of a composite ball structure, in accordance with an embodiment of the present invention;

(34) FIG. **33** is a top perspective view of a clip for ornamental pieces, in accordance with an embodiment of the present invention;

(35) FIG. **34** is a front elevational view of a clip for ornamental pieces, in accordance with an embodiment of the present invention;

(36) FIG. **35** is a side elevational view of a clip for ornamental pieces, in accordance with an embodiment of the present invention;

(37) FIG. **36** is a sketch showing various building kit components, including those of FIGS. **1-35**, in accordance with an embodiment of the present invention; and

(38) FIG. **37** is an alternate sketch showing various building kit components, including those of FIGS. **1-35**, in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(39) The following disclosure is presented to provide an illustration of the general principles of the present invention and is not meant to limit, in any way, the inventive concepts contained herein.

Moreover, the particular features described in this section can be used in combination with the other described features in each of the multitude of possible permutations and combinations

contained herein.

(40) All terms defined herein should be afforded their broadest possible interpretation, including any implied meanings as dictated by a reading of the specification as well as any words that a person having skill in the art and/or a dictionary, treatise, or similar authority would assign thereto.

(41) Further, it should be noted that, as recited herein, the singular forms “a”, “an”, “the”, and “one” include the plural referents unless otherwise stated. Additionally, the terms “comprises” and “comprising” when used herein specify that certain features are present in that embodiment; however, this phrase should not be interpreted to preclude the presence or addition of additional steps, operations, features, components, and/or groups thereof.

(42) All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the principles of the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting principles, aspects, and embodiments of the invention, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future, i.e., any elements developed that perform the same function, regardless of structure.

(43) Reference will now be made to several embodiments of the present invention(s), examples of which are illustrated in the accompanying figures. Wherever practicable, similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

(44) Toy building kits are generally most valuable for the user when they enable him or her the largest breadth of possible building options. This can be assessed both in terms of the kit offering more possible interactions of parts or, more broadly, in terms of the kit enabling a larger variety of constructed structures. The present invention is a refinement towards that end. Possible model forts which can be built are shown conceptually FIG. 1. The flags and/or panel pieces can be coupled to individual connector hubs or poles in the same way that the connector hub pieces are coupled to other individual pieces as described below, or they can be attachable via other, independent mechanisms, such as the clip **610** depicted in FIGS. 33-35, which has levers **612** adapted to widen a mouth **614** of clip **610** when pushed towards each other. Clip **610** can, for example clip onto a pole connector **8**. In any event, these ornamental pieces do not alter the functionality of the remainder of the kit components.

(45) As can be seen in FIGS. 1-3, the majority of a completed structure is defined by a series of pole-shaped connectors **8** (see FIGS. 4-9). These poles **8** define a shape (e.g., a facsimile of a building) by connecting to one another via connector hub pieces. These hubs can be used to line up two poles in the same direction (i.e., collinearly), or the hubs can act as joints to position two poles at angles (e.g., perpendicularly) with respect to one another.

(46) The kit components described herein can be made of plastic or any other suitable material. In certain embodiments, the components can be manufactured via an injection molding process.

(47) In a preferred embodiment, the pole connectors are rigid; however, in some embodiments, they may be flexible. On the most basic level, the poles interact with connector hub pieces by being received in one or more ports on the hub. In some embodiments, for better compatibility with the ports, the ends of the poles may have associated retention devices (not shown), such as a structure at the end of the poles, to facilitate a more secure fit with the ports. These retention devices can be magnetically-based (e.g., in cooperation with magnets in the hubs), use male-to female connectors/threads, or otherwise mechanically secure the pole connectors to the connector hub pieces. In certain embodiments, the retention devices can be selectively removed by the user. In

some embodiments, no retention device is used, and a compression and/or friction fit is relied upon. To this end, poles with larger diameters can enhance the fit between the ports and the pole connectors. To increase building versatility, poles of various sizes and/or lengths can be included in a single kit (see especially FIGS. **36** and **37**). For aesthetic reasons, or to inhibit entry of foreign substances into the hollow diameter of pole connector **8**, an internal shape **5** extending through the length of the pole connector **8** could take up space in the otherwise hollow pole. This internal shape **5** can for instance, be an “X” shape or a “Y” shape and formed by extrusion of material during the manufacturing process of pole connector **8**.

(48) FIG. **10** shows a schematic of a kit component. Specifically, it is a multi-sided five-point connector hub **10** with five lateral ports **12** spaced around the circumference of a cylindrical base **14**. The base **14** itself has a pair of opposing ports **16**, **18** on opposing ends of base **14**. Like the lateral ports **12**, the opposing ports **16**, **18** operate to receive pole connectors. In one embodiment, each opposing port **16**, **18** has a dead-end configuration, wherein there is no interconnection between ports **16**, **18**. In alternate embodiments, a lumen **20** is defined therebetween. In some embodiments, lumen **20** can enable a pole connector (not shown) to pass therethrough, with or without resistance. In another embodiment, one or more of opposing ports **16**, **18** has a stop **22**, which serves to allow ports **16**, **18** to receive a pole connector without allowing the pole to pass completely through lumen **20**. In this manner, two pole connectors can be received, one in each of ports **16**, **18** without any mutual interference. For an improved friction fit, fingers **50** can be provided on the inner walls of ports **12**, **16**, **18**.

(49) FIGS. **16-18** show a representation of a second kit component, a six-point connector hub **110**, which operates similarly to five-point hub **10**. Specifically, six-point hub **110** is multi-sided and has six lateral ports **112** spaced around the circumference of a cylindrical base **114**. The base **114** itself has a pair of opposing ports **116**, **118** on opposing ends of base **114**. Like the lateral ports **112**, the opposing ports **116**, **118** operate to receive pole connectors. In one embodiment, each opposing port **116**, **118** has a dead-end configuration, wherein there is no interconnection between ports **116**, **118**. In alternate embodiments, a lumen **120** is defined therebetween. In some embodiments, lumen **120** can enable a pole connector (not shown) to pass therethrough, with or without resistance. In another embodiment, one or more of opposing ports **116**, **118** has a stop **122**, which enables ports **116**, **118** to receive a pole connector without allowing it to pass through lumen **120**. In this manner, two pole connectors can be received, one in each of ports **116**, **118** without any mutual interference. For an improved friction fit, fingers **150** can be provided on the inner walls of ports **112**, **116**, **118**. Another possibility to improve a friction fit would be to provide slits (not shown) in the walls of ports **116**, **118**, or on the various other ports of the present invention.

(50) In an embodiment, both connectors **10** and **110** can be made of a flexible material (see FIGS. **11**, **13** and **15**). In an embodiment, connectors **10**, **110** are manufactured via an injection molding process. When made to be flexible, lateral ports **12**, **112** can be elastically moved into positions different from their default orientations. Two or more lateral ports **12**, **112** can be moved in the same direction; alternatively, some of ports **12**, **112** can be moved in different (i.e., opposite) directions. In such instances, any lateral ports **12**, **112** positioned in this fashion will flex back to their default position when the forces on them are released. However, when the lateral ports **12**, **112** are connected to a pole connector that is fixed in position (e.g., one that is part of a constructed structure), tension forces will function to maintain lateral ports **12**, **112** in their alternate positions. To this end, hubs **10**, **110** can operate as unique joint connectors (e.g., corner connectors) that can place pole connectors in additional orientations/angles than those described elsewhere herein. In an embodiment, flexible hubs **10**, **110** can be adapted to better comply with the lateral ports **12**, **112** in flexed position. To this end, a plurality of dimples **24**, **124** can be placed around opposing ports **16**, **116**, **18**, **118**. These dimples **24**, **124** serve as a stop to flexed ports **12**, **112** and add to the versatility of hubs **10**, **110**. In an embodiment, the dimples **24**, **124** can be substituted for triangular tabs (not shown) that perform the same function as dimples **24**, **124** (i.e., as a stop) or can complement them.

(51) To improve the ease of connecting and disconnecting poles **8** to hubs **10**, **110** ribbing **26**, **126** can be added to the outer circumference of lateral ports **12**, **112** to facilitate the gripping of ports **12**, **112**. To enable ease of use, ports **12**, **112** can also have a lip **28**, **128** with a diameter greater than the inner diameter of ports **12**, **112**. This adaption facilitates separation of connector poles from hubs **10**, **110**.

(52) In further embodiments of the present invention, additional connectors are provided as part of a kit. FIGS. **19-22** show an eight-sided connector hub **210**. Like connectors **10**, **110**, eight-sided connector hub **210** has a plurality of lateral ports **212** that can receive connector pole(s). Connector **210** is annular in shape, with a central opening **214** passing therethrough. Within said opening **214** is a shelf **216**, a neck **218** below and opposite shelf **216**, and a pair of grooved walls **220** positioned to the sides of shelf **216** and neck **218**, all of which components define opening **214**. The function of these parts will be described hereinbelow. When used with only the pole connectors, connector hub **210** allows for placement of up to eight co-planar poles. For an improved friction fit, fingers **250** can be provided on the inner walls of ports **212**.

(53) FIGS. **23-25** show a two-point connector hub **310**. In an embodiment, hub **310** is cylindrical in shape and has a pair of end ports **312**, **314** which can receive pole connectors. The end ports **312**, **314** can be dead end, or they can be interconnected. Extending from approximately the center of connector **310** is a pair of tabs **316**. Each tab **316** has a pair of slots **318**. In an embodiment of the present invention, two-point connector hub **310** is configured to dock inside central opening **214** of connector **210**. In an unlocked configuration, tabs **316** sit on neck **218**, which functions to prevent connector hub **310** from falling through connector hub **210**. Connector **310** can then be locked into place within central opening **214** by rotating two-point connector hub **310** clockwise or anti-clockwise such that tabs **316** are in close proximity to grooved walls **220**. When locked in place, connector **310** will be prevented from falling out in either direction by shelf **216** in one direction and neck **218** in the other. This first compound structure **320** can be seen in FIG. **30**.

(54) In some embodiments, slots **318** may interact with grooved walls **220** to secure the relative positions of connector hubs **210**, **310**. In further embodiments, preferred directions of rotation can be established via stops **222** on hub **210**, which operate as impediments that prevent the rotation of connector hub **310** by catching tabs **316**. For instance, stops **222** can prevent clockwise or anticlockwise rotation beyond the intended locked and unlocked configurations. Numerous other mechanical means for securing locked and unlocked positions can be implemented, beyond the enumerated examples. For an improved friction fit, fingers **350** can be provided on the inner walls of ports **312**, **314**.

(55) In an embodiment two-point hub **310** can be provided with a window **322**. This window can provide a user with visibility into the interior of two-point hub **310**. This could be useful, for instance, to allow a user to see if an inserted pole (not shown) is fully inserted in end ports **312**. Alternatively, window **322** can be gripped by spring fingers or mechanisms provided on another part to be connected to two-point hub **310** (e.g., when connected to the four-point hub described below). Another adaptation of two-point hub **310** to better interface with another part having fingers is the provision of a recessed area **324** on the exterior of two-point hub **310**. A connecting mechanism can naturally occupy this space for a better fit and easier removal of connected parts.

(56) When connector hubs **210** and **310** are in a locked position, the resulting compound structure **320** has the benefit of both lateral ports **212** and end ports **312**, **314** perpendicular thereto. This exemplary compound hub structure therefore offers a total of ten different directions that connector poles can be extended from and lodged in.

(57) FIGS. **26-29** show an additional, four-point connector hub **410** that can be provided, which can operate as an adapter. Hub **410** has angled ports **412** and a generally hemispherical shape. Connector **410** has a channel **414** passing therethrough. In isolation, connector hub **410** can be used to enable four different orientations of connector poles. In certain embodiments, channel **414** can receive a pole, providing a fifth port. For an improved friction fit, fingers **450** can be provided on

the inner walls of ports **412** and/or **414**. Particularly, spring fingers **416** in channel **414** can facilitate connection with two-point hub **310** by gripping recessed area **324**.

(58) In one embodiment, channel **414** can allow for an end port **312** of connector hub **310** to pass therethrough, forming an intermediate compound structure (not shown). The intermediate compound structure can be configured such that port **312** can still receive a pole connector while in such a configuration. Various modifications (e.g., ridges, slots, grooves, depressions, etc.) can be provided on connectors **310**, **410** to resist relative moment therebetween or to otherwise secure connectors **310**, **410** together; though in some embodiments, rotation of connector **410** about connector **310** may be desirable. In combination with connector **310**, connector **410** effectively gains ports **312**, **314** in addition to angled ports **412**. Furthermore, while connected to connector **410**, connector **310** can further interface with connector **210** as described hereinabove, forming a second compound structure **420** (see FIG. 31). Such a configuration would result in the provision of lateral ports **212** (i.e., eight more ports) to the compound structure hub **420**.

(59) To maximize interconnective possibilities, another hemispherical connector hub **410'** can be added to structure **420**. This would be implemented by receiving the other port **314**, in channel **410'** of the second hemispherical connector **410'**. The resulting ball-shaped compound connector hub **510** (see FIG. 32) therefore has a total of sixteen (or in some embodiments, eighteen) different ports and directions in which it can receive a pole connector. In certain embodiments, angled ports **412'** can be symmetrical with respect to angled ports **412**. Alternatively, angled ports **412** can be anti-symmetrical with respect to angled ports **412'**, which would offer further customizability in terms of possible connections and constructed structures. In alternate embodiments, connector **210** can be omitted from connector hub **510**, resulting in a third compound structure (not shown) of connector hubs **310**, **410** and **410'** that lacks lateral ports **212**.

(60) In certain embodiments, connector hub **410** can be positioned on connector hub **310** in eight different positions (e.g., by rotating connector hub **410** in 45 degree increments). Implemented as such, the compound structures involving hubs **310**, **410** would therefore be able to provide additional angles. In an embodiment, by rotating hemispherical connector **410** and/or hemispherical connector **410'**, up to eight alternate angles could be achieved for a total of twenty-six possible orientations.

(61) It will be understood that the embodiments described herein are merely exemplary and that a person skilled in the art may make many variations and modifications without departing from the spirit and scope of the invention. All such variations and modifications are intended to be included within the scope of the invention.

Claims

1. A toy fort structure building kit including poles and an interconnection configured for receiving the poles, the interconnection comprising: a ring-shaped hub having a lumen passing therethrough and a first plurality of ports disposed along an outer circumference of said ring-shaped hub; a first multi-point hub piece having a cylindrical body and a plurality of tabs extending colinearly from a central section, said first multi-point hub piece being configured to fit into said lumen, and said first multi-point hub piece being lockable with said ring-shaped hub by interfacing with said lumen, wherein said cylindrical body includes a top port at one end of said cylindrical body, and a bottom port at another end of said cylindrical body, opposite said top port; and a second multi-point hub piece having an outer surface having a second plurality of ports oriented away from said outer surface and a central port, said central port being adapted to receive said top port or said bottom port.
2. The kit of claim 1, wherein said first plurality of ports consists of eight ports.
3. The kit of claim 1, wherein said lumen is centrally placed in said ring-shaped hub.
4. The kit of claim 1, wherein said first plurality of ports include dead-end geometries.

5. The kit of claim 1, wherein said first multi-point hub piece has a cylindrical body.
 6. The kit of claim 1, further comprising a two point hub piece, wherein said two-point hub piece and said first multi-point hub piece are configured to become interlocked to form a composite structure.
 7. The kit of claim 6, wherein said ring-shaped hub piece further comprises a shelf coupled to said ring shaped-hub piece and located at one end of said lumen, said shelf having a slot with a narrower diameter than the diameter of said lumen.
 8. The kit of claim 7, wherein said composite structure is formed when said tabs are engaged with said shelf.
 9. The kit of claim 8, wherein said first multi-point hub piece is rotatable while in said lumen of said ring-shaped hub such that said first multi-point hub piece can be selectively locked and unlocked into place within said lumen through clockwise and/or anticlockwise rotation, whereby said tabs can be selectively engaged or disengaged with said shelf.
 10. The kit of claim 1, wherein said second plurality of ports include dead-end geometries.
 11. The kit of claim 1, wherein said second multi-point hub piece is hemispherical in shape.
 12. The kit of claim 11, wherein said composite structure can receive said second multi-point hub piece at said top port and said bottom port simultaneously, thereby forming a ball structure.
 13. The kit of claim 12, wherein said ball structure has a total of sixteen different ports.
 14. The kit of claim 1, wherein said second plurality of ports consists of four ports.
 15. The kit of claim 14, wherein each port of said second plurality of ports is angled away from a top surface of said second multi-point hub piece.
 16. The kit of claim 15, wherein said second multi-point hub piece includes teeth within said central port, said teeth adapted to interface with a recess localized on said first multi-point hub piece.
 17. The kit of claim 1, further comprising a flexible hub piece which includes a plurality of flexible ports.
 18. The kit of claim 17, wherein said plurality of flexible ports consists of five or six ports.
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